

R&T



Cours Réseaux : Switched Port Analyzer (SPAN)

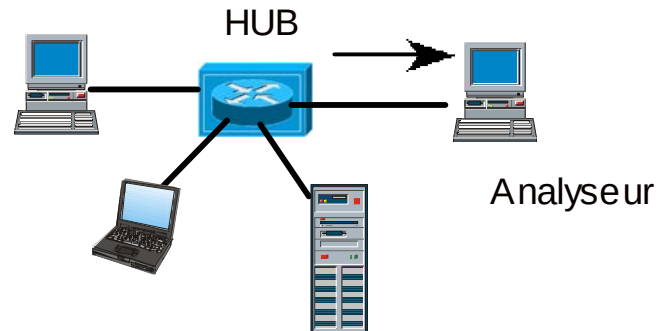
SPAN

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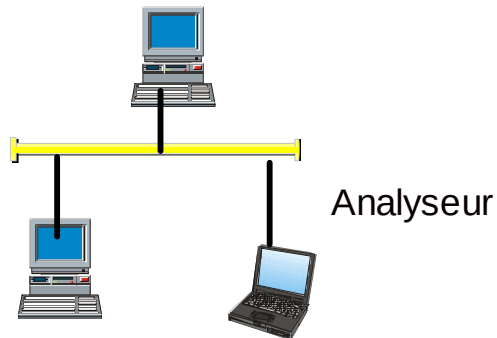
- Introduction
- Principes
- Configurations SPAN & RSPAN
- Vérifications
- Conclusion

Introduction(1)

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Modèle "Broadcast"



Analyse de trames :

- Hub
- Mode promiscuous

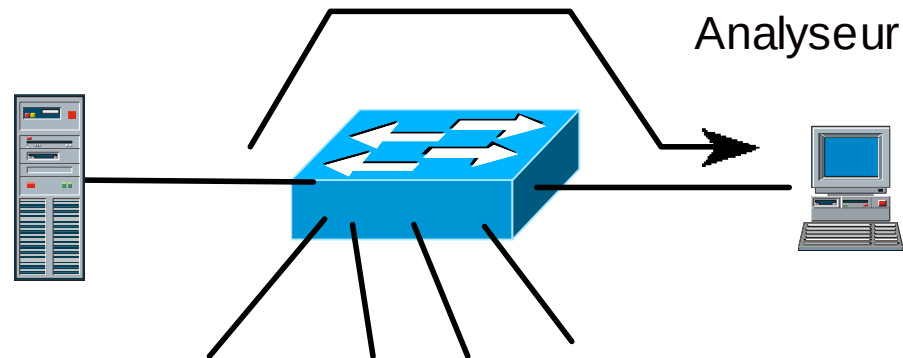
No.	Time	Source	Destination	Protocol	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	DHCP Discover - Transaction ID 0x611ca31b
2	0.007990	192.168.37.1	192.168.37.3	DHCP	DHCP Offer - Transaction ID 0x611ca31b
3	0.023609	0.0.0.0	255.255.255.255	DHCP	DHCP Request - Transaction ID 0x611ca31b
4	0.031527	192.168.37.1	192.168.37.3	DHCP	DHCP ACK - Transaction ID 0x611ca31b
5	0.036872	00:0d:54:9c:4d:5d	ff:ff:ff:ff:ff:ff	ARP	Gratuitous ARP for 192.168.37.3 (Request)
6	0.684875	00:0d:54:9c:4d:5d	ff:ff:ff:ff:ff:ff	ARP	Gratuitous ARP for 192.168.37.3 (Request)
7	1.686321	00:0d:54:9c:4d:5d	ff:ff:ff:ff:ff:ff	ARP	Gratuitous ARP for 192.168.37.3 (Request)

Environnement commuté

- SPAN

SPAN:Principes

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SPAN

- Switched Port Analyzer
- Port Miroir du trafic
- 2 ports SPAN max par Switch



RSPAN

- Remote SPAN
- Miroir vers un port distant

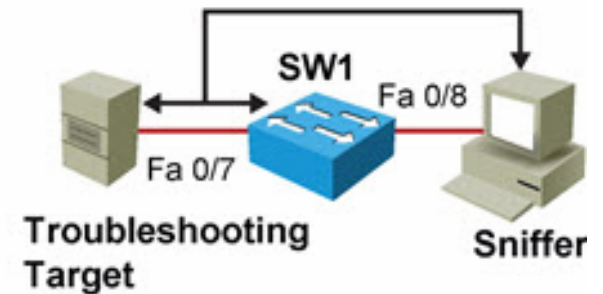
SPAN : Configuration

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```
monitor session 1 source interface Fa0/7  
monitor session 1 destination interface Fa0/8
```

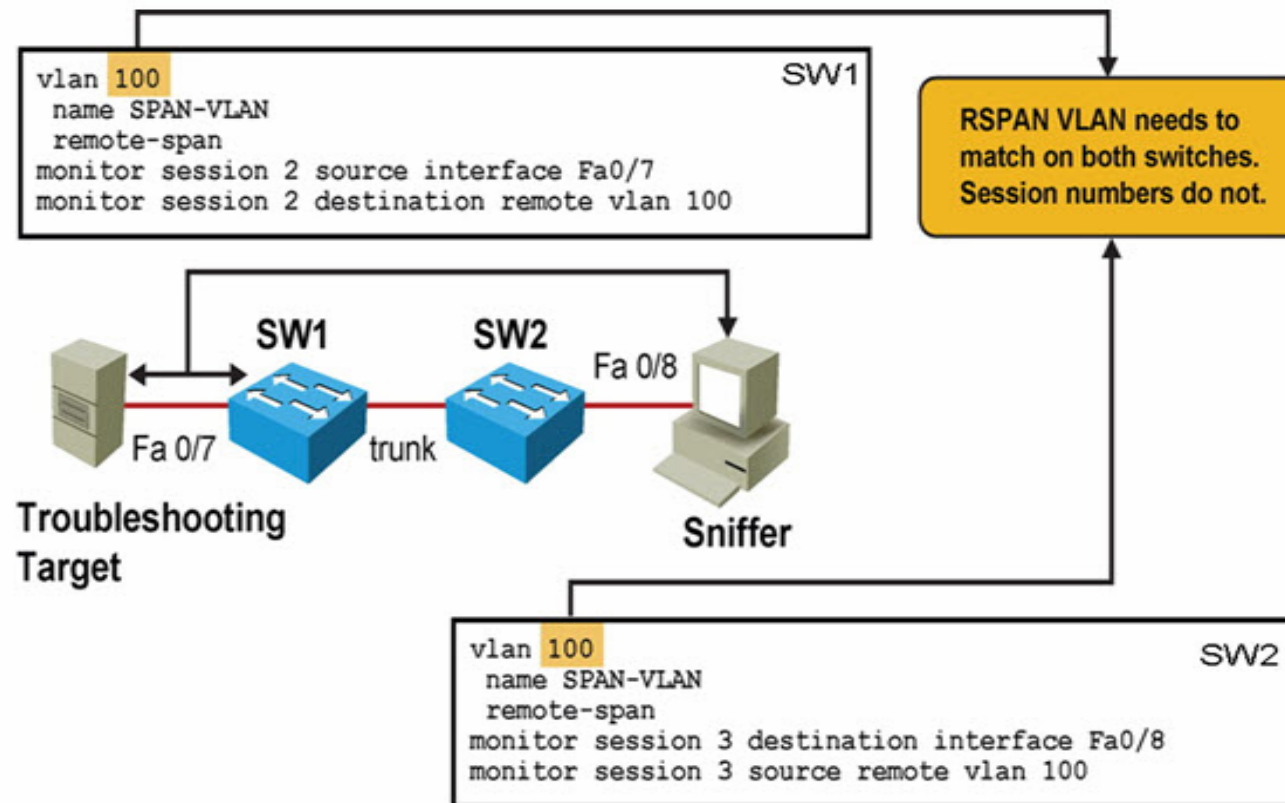
Sources and destinations that form a single SPAN session are identified by a session number

```
SW1#show monitor  
Session 1  
-----  
Type : Local Session  
Source Ports :  
    Both : Fa0/7  
Destination Ports : Fa0/8  
Encapsulation : Native  
Ingress : Disabled
```



RSPAN : Configuration

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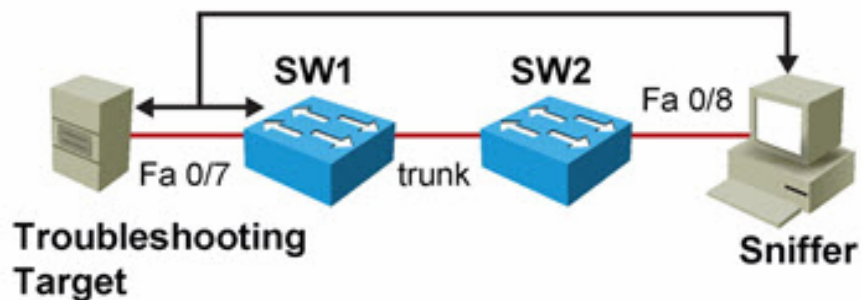


Configuration

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```
SW1#show monitor
Session 2
-----
Type                : Remote Source Session
Source Ports        :
    Both             : Fa0/7
Dest RSPAN VLAN     : 100
```

```
SW1#show vlan remote-span
Remote SPAN VLANs
-----
100
```



```
SW2#show vlan remote-span
Remote SPAN VLANs
-----
100
```

```
SW2#show monitor
Session 3
-----
Type                : Remote Destination Session
Source RSPAN VLAN   : 100
Destination Ports    : Fa0/8
Encapsulation        : Native
Ingress              : Disabled
```

Conclusion

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- SPAN/RSPAN nécessaire pour l'analyse de protocoles
- Très utile pour l'administration de réseaux.
- Approche par la pratique : TPs!!!